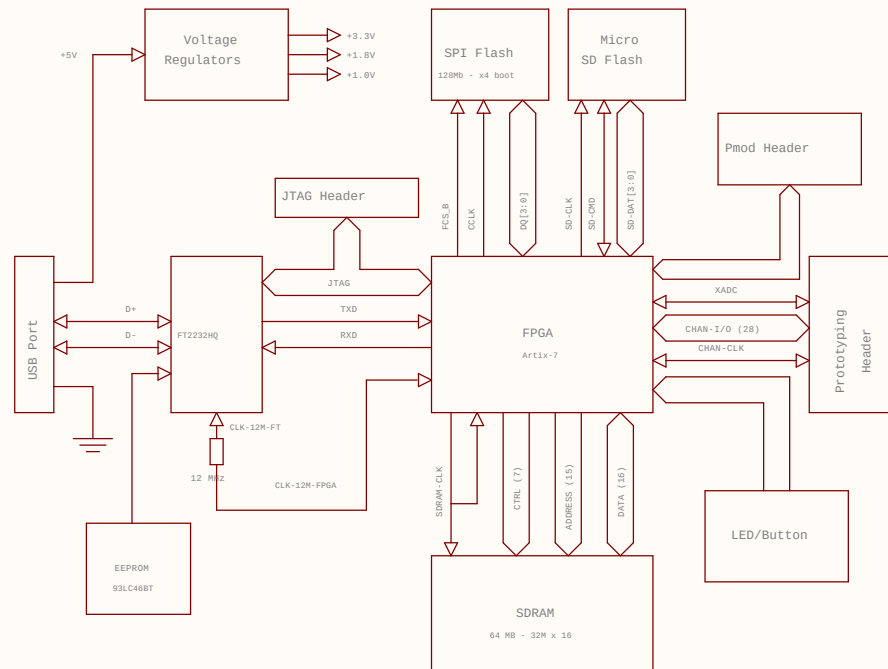


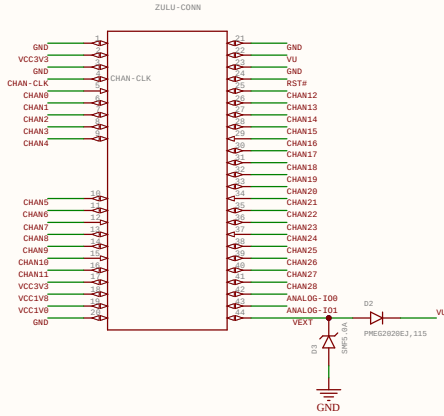
SPI flash sits on the FPGA's dedicated master-SPI config pins: CCLK C11, FCS_B K19, DQ[3:0] on D18/D19/G18/F18. Read at power-up.

microSD runs the native 4-bit SD bus, not SPI: CLK, CMD and DAT[3:0] on six bank-34 balls of its own, so a card cannot touch the config bus.

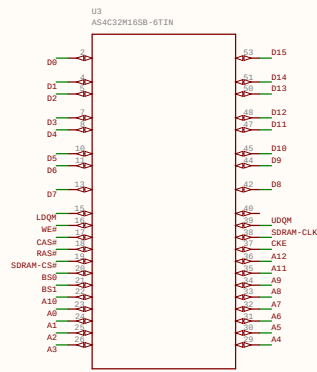


Block Diagram			
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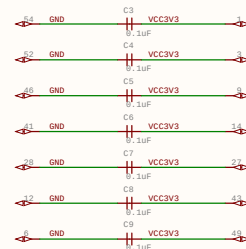
X2: 44 pins on a 2.54 mm grid, rows 0.700 in apart, board 69.85 x 20.32 mm.
Numbered right to left and straight through: top row 1-20, bottom row 21-44.
TOP ROW, x 50.69 down to 1.27: GND, +3.3V, GND, then CHAN-CLK and CHAN0..CHAN3
(pins 4-8), CHAN4 (pin 9), CHAN5..CHAN11 (10-16), +3.3V, +1.8V, +1.8V, GND.
FOUR positions between pin 9 and pin 10 -- x 36.83, 34.29, 31.75, 29.21 -- are
RESERVED, not spare: the landing for the edge-mounted micro-USB X1, 7.80 mm of
copper in 11.176 mm clear, centred with 1.69 mm each side.
BOTTOM ROW, contiguous, x 50.69 down to 1.27: GND, VU, GND, RST#, CHAN12..CHAN28
(pins 25-41), ANALOG-IO0, ANALOG-IO1, +5V-INPUT.
FIVE grounds: pins 1, 9, 20, 21, 23. Pin 22-VU SOURCES 5 V, pin 44 only takes it.



U3 = AS4C32M16SB-6T1N: 512Mbit (32M x 16) = 64MB.
Pin-for-pin and dimensionally identical to the 256MB
AS4C16M16A-6T1N it replaces -- all 54 pins and every
TSP: 1 dimension checked against both datasheets, and
pin 40 is NC on both.
ONE DIFFERENCE, AND IT IS NOT A PIN: the column address
is A0-A9 (1024 columns), not A0-A8 (512). The memory
controller must be set to 10 column bits, or it reaches
only half the array. Row A0-A12, banks BA0/BA1 and
refresh (8192 cycles / 64ms) are unchanged.
IDD1 rises 60mA -> 120mA, on the 600mA VCC3V3 buck
(LTC3569 SW3) -- worth a rail budget check.



U3 DECOUPLING -- each cap bridges one supply pin to the ground pin OPPOSITE it.
In the 54-pin TSOP-17, pin n faces pin 55-n and every supply pin faces the ground of its own domain: VDD 1/34/21 to VSS 54/41/28, VDDQ 5/9/43/49 to VSSQ 52/46/12/7. Alliance isolates VDDQ from VDD for DQ noise immunity. LAYOUT MUST keep the PAIRS: each cap on the underside spanning its two pins, own via, VDDQ returning to VSSQ and VDD to VSS. C10 is bulk and pairs with nothing. DO NOT split the ground plane.



Memory			
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